MINISTRY OF ECONOMY
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DOSM DATATHON

DOSM Datathon 2026

"Sustainable Tourism Readiness in Malaysia"

by

Dataverse

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INTRODUCTION

a. Background of Study

Tourism is a significant contributor to Malaysia's economy as it supports economic activity, employment, and tourism-related expenditure which includes the total money spent by visitors on accommodation, food, transport, shopping, and activities during their travel and stay. In 2023, Malaysia's tourism industry generated RM275.8 billion and constituted 15.1% of the country's Gross Domestic Product (GDP). Moreover, the tourism industry also employed approximately 3.4 million people, making up 21.4% of the total labor force (DOSM,2024). Besides that, domestic tourism also recorded significant growth with 213.7 domestic visitors and RM84.9 billion in domestic expenditure in 2023 (DOSM,2024).

On the other hand, tourism development cannot be assessed entirely based on visitor numbers and tourism expenditure. The consideration of economic, environmental, and social factors is vital for sustainable tourism in order to ensure that tourism activities can bring benefits while reducing negative impacts on natural resources and local communities at the same time. In Malaysia, tourism activity is interconnected with economic growth, energy consumption, and environmental outcomes (Raihan et al., 2023). This highlights the importance of taking into account multiple factors to evaluate sustainable tourism development.

Moving on, the social dimension is also crucial because tourism development directly affects the communities in destination areas. According to the research conducted in Malaysia's Lenggong Valley World Heritage Site, the residents' perceptions influenced their support and participation in sustainable tourism development (Rasoolimanesh & Jaafar, 2017). Therefore, community-related factors and socioeconomic conditions must also be considered when measuring tourism sustainability.

The accessibility of official statistical data enables the development of a more comprehensive approach to sustainable tourism analysis. Department of Statistics Malaysia (DOSM) offers statistics on tourism, population, employment, income, economic indicators, as well as other socioeconomic indicators. Incorporating these indicators can provide a broader understanding of divergent conditions associated with sustainable tourism across Malaysian states.

On top of that, there is a growing trend in Malaysia of utilizing machine learning algorithms to evaluate tourism sustainability. Predictive analytics and machine learning can be used to investigate sustainable tourism outcomes (Abdul Mohaimin et al.,2025). Besides that, applied supervised machine learning can also be used to classify sustainable tourism practices within

Malaysia's accommodation sector (Abdul Rahim et al., 2026). Thus, studies reflect the potential of machine learning in supporting data-driven tourism analysis.

In conclusion, this project focuses on developing a state-level tourism analytical framework by using official Malaysian statistics across four core pillars that are, economic, social, environmental, and tourism dimensions. By applying unsupervised K-Means clustering and supervised Random Forest classification, the framework identifies states with similar characteristics and isolates critical performance drivers. The findings are finally synthesized through an interactive Power BI dashboard, enabling a data-driven method to comprehend and navigate differences in sustainable tourism conditions across Malaysian states.

b. Problem Statement

Tourism performance is ordinarily measured using indicators such as tourism expenditure, visitor arrivals, number of trips, and length of stay. Even though these indicators are impactful in understanding tourism activity, they do not fully depict the sustainability conditions of a destination. To illustrate, a state might have high tourism activity and face challenges related to socioeconomic conditions or environmental sustainability simultaneously.

The multidimensional nature of sustainable tourism is backed by existing research. Tourism activity in Malaysia has been correlated with economic growth, energy consumption, and environmental outcomes (Raihan et al., 2023). Furthermore, studies of Malaysian tourism destinations have pointed out the significance of environmental protection and the involvement of the community (Rasoolimanesh & Jaafar, 2017). Thus, sustainable tourism assessment should take into account the combination of divergent factors instead of relying only on tourism performance indicators alone.

Conversely, even though official statistics are available, relevant information and insights are mostly distributed across different statistical areas and not necessarily accessible as an integrated state-level sustainable tourism assessment. On top of that, existing Malaysian machine-learning research has mostly focused on specific problems such as predicting tourism outcomes or classifying sustainable practices within accommodation businesses (Abdul Mohaimin et al., 2025; Abdul Rahim et al., 2026).

To conclude, there is an opportunity to develop an integrated analytical framework that uses official Malaysian statistics to study and assess sustainable tourism characteristics across states. This project addresses the issue by combining relevant economic, social,

environmental, and tourism indicators, applying applicable machine learning to measure state-level profiles and project the findings through an interactive dashboard. The proposed framework aims to provide a clear and more comprehensive representation of sustainable tourism conditions across Malaysia.

c. Objectives

The objectives of the project include:

1. To integrate and harmonize official Malaysian economic, social, environmental, and tourism statistics into a unified state-level dataset covering all states and federal territories (reference year 2022).
2. To apply K-means clustering to segment states into distinct readiness profiles, and to use a Random Forest to identify which indicators most distinguish those profiles.
3. To construct a transparent Sustainable Tourism Readiness Index (STRI) by min-max normalizing indicators and aggregating them into four equally weighted dimensions (economic, social, environmental, tourism), producing state rankings and readiness categories.
4. To deliver the results as an interactive dashboard and baseline demand forecast that helps policymakers identify priority states for sustainable tourism intervention.

LITERATURE REVIEW

a. Sustainable Tourism Framework and Key Concepts

Sustainable tourism considers the economic, social, and environmental impacts of tourism on destinations and their communities (United Nations Statistical Commission, 2023). Thus, visitor numbers and expenditure alone do not indicate sustainability. Sustainable tourism therefore requires consideration of economic benefits, environmental protection, and community well-being.

Early composite scoring frameworks such as those by Fernández and Rivero (2009) established that tourism sustainability cannot be represented by a single scalar indicator. However, composite measures require careful decisions regarding indicator selection, normalisation, weighting, and aggregation. In modern regional tourism research, several key sustainability concepts have developed. Jeurissen (2000) discussed Elkington's Triple Bottom Line (TBL) framework, which considers economic, social, and environmental dimensions

through the concepts of profit, people, and planet. Recent tourism research applies this concept to regional sustainability by considering economic, social, and environmental conditions together (Nica et al., 2025).

Secondly, Tourism Carrying Capacity (TCC) traditionally focused on the capacity of destinations to manage tourism pressure (O'Reilly, 1986). Recent spatial studies consider broader regional conditions, including resources, infrastructure, social conditions, and environmental pressure (Wu et al., 2023). This perspective supports the consideration of broader regional conditions when assessing tourism development readiness.

b. Empirical Studies on Regional and Subnational Sustainable Tourism

Previous research has examined sustainable tourism at regional and subnational levels using different statistical and analytical methods. These studies show the importance of considering geographical differences when evaluating tourism sustainability.

Zhu et al. (2023) studied tourism development and environmental pressure in the Yangtze River Basin using data from 2000 to 2019. The researchers developed separate measures for tourism development and environmental pressure and examined the relationship between them. The findings showed clear differences between regions in terms of tourism development and environmental pressure. This highlights the importance of considering regional differences and the environmental effects of tourism development. However, the study mainly focused on economic and environmental factors and gave less attention to social factors. Since the study focused on a specific region in China, its findings may not directly apply to Malaysian states. This supports the need for a broader assessment that considers economic, social, environmental, and tourism factors in Malaysia.

Li et al. (2024) examined the sustainable development of tourism in the Chengdu-Chongqing urban agglomeration. The study examined how tourism sustainability changed over time and identified the factors influencing it. Several indicators were combined to assess tourism sustainability at the regional level. The findings show that using multiple indicators can provide a more complete assessment of tourism sustainability. However, the framework was designed for a specific region in China and may not reflect the economic, social, and environmental conditions of Malaysian states. Therefore, a similar assessment needs to be adapted to the Malaysian context.

Xu and Li (2025) studied sustainable tourism management in coastal destinations in the Bohai Rim using tourism carrying capacity. The study focused on the pressure caused by

tourism growth and the ability of destinations to manage its environmental and social impacts. The findings show that excessive tourism activity can increase environmental pressure and affect local communities. This highlights the importance of considering a destination's capacity before expanding tourism. However, the study mainly focuses on tourism pressure and does not fully consider the wider economic and social conditions of each state.

Hoffmann et al. (2022) examined sustainable tourism measurement using web-scraped data from an online tourism platform and supervised machine learning to predict sustainability labels for accommodation establishments. The study found that platform data could provide cost-effective sustainability information with high spatial and temporal detail. This shows the potential of machine learning for tourism analysis. However, reliance on online platform data may reflect the characteristics of users and businesses represented on the platform. This study uses official OpenDOSM data to provide a consistent statistical foundation for state-level analysis in Malaysia.

The OECD (2024) developed a regional framework for measuring tourism sustainability across Andalusia, Catalonia, Navarra, and the Region of Valencia using economic, social, environmental, and governance indicators. The framework was developed through stakeholder consultation, technical workshops, indicator assessment, and pilot testing. The study highlighted the value of regional indicators for sustainability monitoring while identifying challenges related to data availability and different regional priorities. However, the framework was designed for Spanish regions and may not be directly applicable to Malaysia. Therefore, the present study adapts the regional approach using Malaysian official statistics to develop a state-level analytical framework.

Overall, these studies show that regional tourism sustainability can be assessed using multidimensional indicators and spatially differentiated approaches. However, differences in geographical context, indicators, and analytical methods limit their direct applicability to Malaysian states. This supports the need for a context-specific framework using consistent state-level indicators.

c. Machine Learning and Decision Support Systems

Machine learning is increasingly used to identify patterns in complex sustainability data. K-Means clustering is an unsupervised method that groups similar observations based on selected variables. Schmitz (2024) applied K-Means to sustainability data from 2,485 companies worldwide and showed that the method can identify sustainability patterns without predefined categories. This is relevant to the present study because Malaysian states do not have fixed sustainability categories. Therefore, K-Means can be used to group states based on their economic, social, and environmental characteristics.

Decision support systems combine data, analytical methods, and visual tools to support decision-making. De La Cal et al. (2026) developed a decision support system for sustainable cultural tourism using multiple sustainability indicators and an interactive dashboard. The study shows that dashboards can help stakeholders understand sustainability data and compare different indicators. However, the study focuses on European cultural tourism, which may not fully represent the Malaysian context. Therefore, a similar approach can be adapted to support sustainable tourism planning in Malaysia.

d. Malaysian Context

Recent studies have explored the use of data analytics and machine learning for sustainable tourism in Malaysia. Mohaimin et al. (2025) studied sustainable tourism development in Malaysia using data preparation, exploratory analysis, descriptive and predictive analytics, and feature engineering. The study used Gradient Boosting and Support Vector Machine models to predict sustainable tourism outcomes. The findings show that machine learning can support data-driven tourism decisions. However, the study mainly focuses on prediction and forecasting, while the present study focuses on grouping Malaysian states based on their sustainability characteristics. Therefore, this study uses K-Means clustering to provide a different approach by identifying patterns across multiple economic, social, environmental, and tourism indicators.

e. Research Gap

The literature reveals several gaps in sustainable tourism research. First, previous studies focus on specific destinations, tourism performance, carrying capacity, or environmental pressure, while fewer studies integrate economic, social, and environmental indicators into a single state-level framework. Second, although machine learning has increasingly been applied to tourism sustainability, recent studies often focus on prediction or specialised

datasets. However, limited research has applied unsupervised clustering to Malaysian state-level sustainable tourism data. Third, existing decision-support studies demonstrate the value of interactive dashboards, but there is limited integration of machine learning with official Malaysian state-level economic, social, and environmental indicators for sustainable tourism assessment.

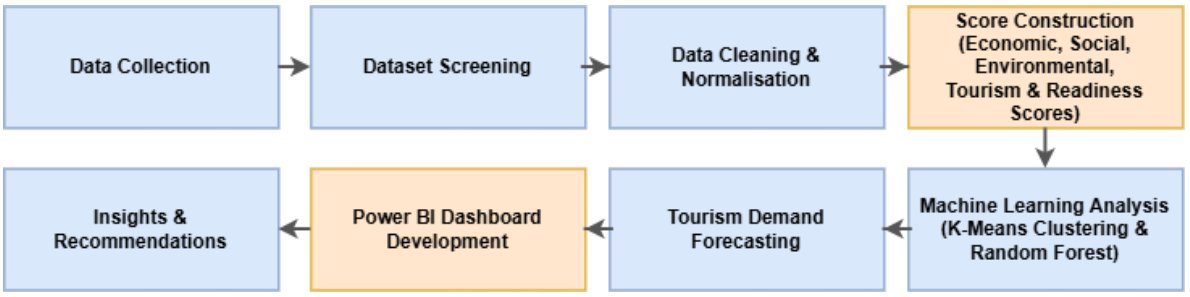
METHODOLOGY

a. Research Framework

This study adopts a data-driven analytical framework to assess Sustainable Tourism Readiness across Malaysian states. The framework integrates official statistical data, machine learning techniques, predictive analytics, and interactive dashboard visualisation to generate evidence-based insights for sustainable tourism planning and policy development.

The methodology consists of six principal stages: data collection, dataset screening, data preprocessing, machine learning analysis, sustainable tourism assessment, and dashboard development. The overall framework is designed to identify tourism readiness patterns, evaluate tourism pressure, forecast tourism demand, and provide decision-support insights for stakeholders.

Research Workflow



b. Data Collection and Preparation

The study utilised official state-level datasets obtained from OpenDOSM and DOSM publications. The datasets covered tourism, economic, social, environmental, demographic, and labour market indicators. The final dataset included all 13 Malaysia states and Federal Territories.

Several datasets identified during the initial planning stage were excluded due to inconsistent reporting periods, incomplete records, outdated observations, or limited relevance to sustainable tourism assessment. Additional suitable datasets were subsequently integrated to improve analytical validity.

Prior to analysis, the datasets underwent preprocessing procedures including duplicate removal, missing value treatment, data integration, and standardisation. Min-Max Normalisation was applied to transform indicators into a comparable 0-100 scale:

$$Score_i = \frac{X_i - X_{min}}{X_{max} - X_{min}} \times 100$$

where Xi represents the indicator value for a state, while Xmin and Xmax represent the minimum and maximum state values respectively.

c. Sustainable Tourism Readiness Framework

To evaluate tourism sustainability comprehensively, four composite indices were developed.

Economic Score

The Economic Score represents economic readiness and was derived from GDP per capita, mean household income, and labour participation rate.

Social Score

The Social Score measures social wellbeing using poverty rate, unemployment rate, and Gini coefficient. Lower poverty, lower unemployment, and lower inequality contribute to a higher social score.

Environment Score

The Environment Score evaluates environmental sustainability using forest reserve area, treated water access, and water consumption indicators.

Tourism Score

The Tourism Score measures tourism performance using tourism intensity, occupancy rate, and international hotel guest arrivals.

Readiness Score

The Sustainable Tourism Readiness Score was calculated by combining the Economic, Social, Environmental, and Tourism Scores to provide an overall assessment of sustainable tourism readiness for each state.

d. Machine Learning Analysis

K-Means Clustering

K-Means Clustering was applied to classify Malaysian states based on similarities in tourism, social, environmental, and economic indicators. The model analysed 13 states and 3 Federal Territories and generated three readiness clusters. Cluster quality was evaluated using the Silhouette Score, which produced a value of 0.31.

Random Forest Feature Importance

Random Forest analysis was conducted to identify the variables contributing most significantly to tourism readiness.

The most influential indicators were:

- Poverty Score (0.17)
- Inequality Score (0.14)
- Labour Participation Rate (0.13)
- Mean Household Income (0.13)
- Median Household Income (0.12)
- Treated Water Access (0.11)

Additional contributing factors included Forest Reserve per Population (0.07), Average Occupancy Rate (0.04), GDP per Capita (0.03), Unemployment Score (0.03), Tourism Guest Score (0.02), and Water Pressure Score (0.02).

The results indicate that social and socioeconomic factors are the strongest determinants of sustainable tourism readiness in Malaysia. In particular, poverty, income distribution, labour market participation, and access to basic infrastructure demonstrated greater influence than tourism activity indicators alone. This finding suggests that sustainable tourism development depends not only on visitor demand and tourism performance, but also on broader social

wellbeing, economic inclusiveness, and environmental sustainability factors.

The complete data preprocessing, K-Means clustering, and Random Forest classification code https://colab.research.google.com/drive/106P5uxguvPyJ07XXjBXBvny5y7noUiFa?usp=sharing#scrollTo=R27tV5PMcYHI

Interpretation of K-Means Results

The K-Means clustering model was developed to identify groups of Malaysian states with similar sustainable tourism characteristics. The optimal number of clusters was determined using the Silhouette Score, which evaluates cluster separation and cohesion. The model identified **three distinct clusters**, indicating that Malaysian states exhibit different levels of sustainable tourism readiness.

States in Cluster 0 generally demonstrated stronger economic performance and higher readiness levels, while Cluster 1 represented emerging states with moderate sustainability characteristics. Cluster 2 contained states facing greater socioeconomic and infrastructure-related challenges. These results provide a clearer understanding of regional disparities and support targeted tourism planning strategies.

Interpretation of Random Forest Results

The Random Forest model was applied to determine which indicators contributed most significantly to sustainable tourism readiness. Feature importance analysis revealed that **poverty score (0.17)**, **inequality score (0.14)**, **labour participation rate (0.13)**, **mean household income (0.13)**, **median household income (0.12)**, and **treated water access (0.11)** were the strongest predictors of readiness.

The findings indicate that sustainable tourism readiness is influenced primarily by socioeconomic wellbeing and access to essential infrastructure rather than tourism activity alone. This suggests that states seeking to strengthen tourism sustainability should focus on reducing poverty, improving income distribution, increasing workforce participation, and enhancing access to basic public services.

e. Tourism Forecasting and Dashboard Development

Historical tourist arrival data from 2020 to 2024 were analysed to generate future tourism demand projections. The forecasting outputs were integrated into the dashboard through

Actual Arrivals, Next Month Forecast, and Year-over-Year Growth indicators.

The final Power BI dashboard consists of three analytical modules:

1. **Tourism Demand Forecast** – tourism trend and forecasting analysis.
2. **State Tourism Pressure** – tourism intensity, occupancy rate, and international guest analysis.
3. **Sustainability Analysis** – readiness scores, cluster classifications, and sustainability indicators.

The dashboard functions as an interactive decision-support system for policymakers, tourism stakeholders, and researchers.

Interpretation of Tourism Forecasting Results

The forecasting model was developed using historical monthly tourist arrival data from 2020 to 2024 to estimate future tourism demand patterns. A seasonal baseline approach was applied to identify recurring monthly tourism trends and generate forecasts for subsequent periods.

The forecast results indicate that Malaysia's tourism sector is expected to maintain a positive recovery trajectory following the post-pandemic rebound observed from 2022 onwards. Historical records show a substantial increase in tourist arrivals between 2022 and 2024, while the forecasted values suggest continued visitor demand throughout the projected period.

The forecasting module provides three key indicators:

- **Latest Actual Arrivals:** 3.10 million visitors.
- **Next Month Forecast:** 2.36 million projected visitors.
- **Year-on-Year Growth:** -0.16%.

These indicators enable tourism stakeholders to anticipate future demand fluctuations and support evidence-based planning, resource allocation, and destination management decisions. The forecasting outputs also complement the sustainability assessment by providing a forward-looking perspective on tourism development.

f. Alignment with Sustainable Development Goals (SDGs)

The proposed framework aligns directly with the DOSM Datathon 2026 SDGs:

1. **SDG 8: Decent Work and Economic Growth** through employment, income, and tourism economic indicators.
2. **SDG 12: Responsible Consumption and Production** through tourism pressure and sustainability assessment.

FINDINGS

a. Information Architecture and Solution Approach

The proposed solution is structured around three integrated layers that transform raw statistical data into decision-support insights.

The data integration layer consolidates official statistics from OpenDOSM, Tourism Malaysia's Paid Accommodation Survey, and data.gov.my's monthly arrivals-by-state-of-entry records into a single, harmonised state-level master dataset. This stage involved dataset screening (e.g., excluding **domestic_hotel_guests** due to incomplete 2015–2018 coverage, and restricting entry-point arrival data to national-level aggregation only, since it disproportionately reflects Johor's Singapore land border rather than genuine state-level visitation) to ensure analytical validity.

The analytical layer applies Min-Max normalisation to standardise all indicators to a comparable 0–100 scale, from which four composite dimension scores (Economic, Social, Environmental, Tourism) and an overall Readiness Score are constructed. This layer also incorporates two machine learning components: K-Means clustering, which segments the 16 states/federal territories into readiness profiles without a predefined target variable, and Random Forest classification, which quantifies the relative importance of each indicator in distinguishing those profiles.

The presentation layer operationalises these outputs into an interactive Power BI dashboard comprising three linked analytical modules — Demand Forecast, State Tourism Pressure, and Sustainability Analysis — connected through a relational data model **(Master ↔ State_Tourism_Pressure** on a 1:1 bidirectional relationship, and **Master ↔ ML_Cluster_Profile** on a many-to-one relationship via the **cluster** field). This architecture allows a policymaker to move fluidly from a national demand trend, to a specific state's tourism pressure, to that state's underlying sustainability and cluster profile, within a single connected interface.

b. Results: Sustainable Tourism Readiness Scores

The Sustainable Tourism Readiness Score ranged from 23.49 (Kelantan) to 76.23 (W.P. Kuala Lumpur) across the 16 states and federal territories, with a national average of 46.91. As shown in

Figure 7 below, the three highest-readiness states — W.P. Kuala Lumpur, W.P. Putrajaya, and Selangor — are all federal-territory or highly urbanised administrative centres, while the three lowest-readiness states — Kelantan, Sabah, and Kedah — are geographically peripheral states. This distribution suggests a pronounced urban-periphery divide in sustainable tourism readiness, consistent with broader socioeconomic disparities documented in Malaysia's regional development literature.

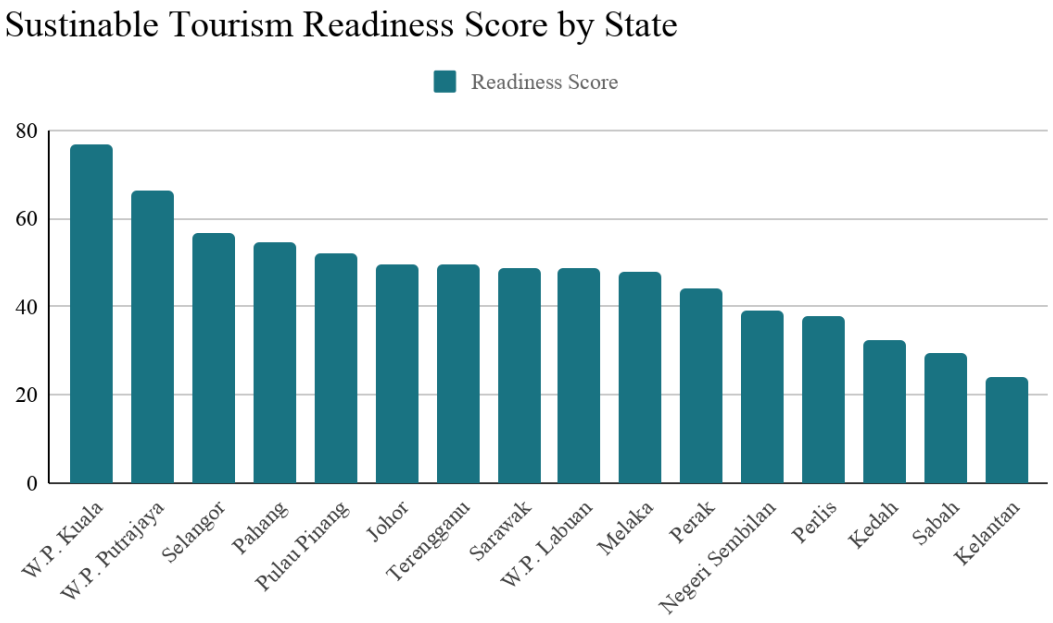


Figure 7 *Sustainable Tourism Readiness Score by State*

c. Results: Tourism Pressure and Demand Patterns

The State Tourism Pressure module recorded a national average occupancy rate of 46.26% and average tourism intensity (international guests per capita) of 440.82, with total international hotel guests of approximately 14 million (2015–2022). Sarawak, Terengganu, and Perak recorded the highest occupancy rates, while W.P. Kuala Lumpur, Johor, and Pahang recorded the highest absolute international guest volumes. Notably, states with high guest volume do not always correspond to the highest occupancy or tourism-intensity rankings, confirming that raw visitor volume and per-capita tourism pressure represent distinct dimensions of tourism performance — a distinction a single "tourist arrivals" indicator would have obscured. This validates the project's methodological decision to use population-adjusted tourism intensity, rather than raw counts, as the basis for the Tourism Score.

d. Machine Learning Insights: Cluster Profiles

K-Means clustering (k = 3, silhouette score = 0.309) grouped the 16 states into three distinct readiness profiles:

- **High Readiness (Cluster 0) – Urban Economic Hubs** (W.P. Kuala Lumpur, W.P. Putrajaya, Selangor; n=3): the highest GDP per capita (RM71,839), highest mean/median income, strongest labour participation (75.73%), near-universal treated water access (99.63%), and by far the highest tourism guest score (963.29) and occupancy rate (51.3%). However, it also showed the most severe water pressure score, indicating that high urban density and tourism activity place the greatest strain on water resources.
- **Moderate Readiness (Cluster 1) – Mixed-Development States** (Johor, Pulau Pinang, Kedah, Melaka, Negeri Sembilan, W.P. Labuan, Perak, Pahang, Perlis, Terengganu; n=10): the largest and most heterogeneous group, with mid-range GDP per capita (RM41,609), moderate poverty/unemployment scores, and moderate tourism intensity (guest score 369.48, occupancy 43.72%). Readiness Scores within this group span a wide range (32.18–54.07).
- **Emerging Readiness (Cluster 2) – Resource-Rich, Economically Lagging States** (Kelantan, Sabah, Sarawak; n=3): the lowest GDP per capita (RM31,880), most severe poverty and unemployment scores, and lowest treated water access (79.37%) — but by far the highest forest-per-population ratio. Despite the lowest tourism guest volume (156.14), this group's occupancy rate (49.67%) exceeded that of the Moderate Readiness group, suggesting smaller-scale but proportionally active accommodation demand.

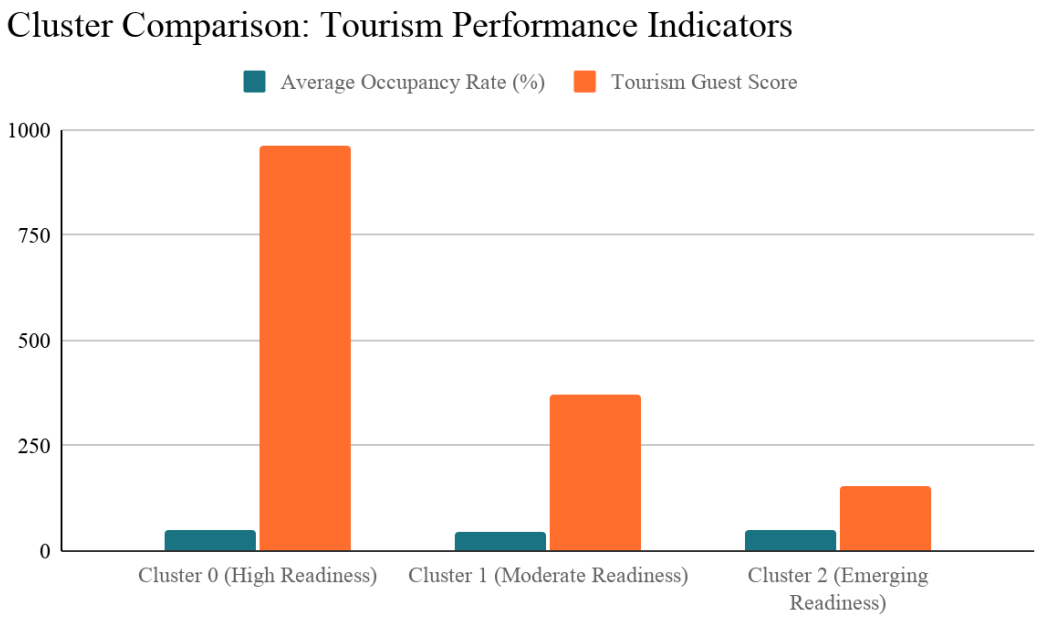


Figure 8 *Cluster Comparison: Tourism Performance Indicators*

Random Forest feature importance analysis (across all clustering features) identified Poverty Score (17.0%), Median Income (12.0%), Inequality Score (14.0%), Treated Water Access (11.0%), Mean Income (13.0%), and Labour Participation Rate (13.0%) as the most influential predictors of cluster membership, while tourism-specific indicators ranked comparatively low. This is a central finding of the project: **sustainable tourism readiness in Malaysia is driven predominantly by underlying socioeconomic conditions rather than tourism activity itself.**

e. Interpretation

The clustering results reinforce the Random Forest finding that socioeconomic indicators — rather than tourism activity — primarily separate high- and low-readiness states. Cluster 0's water pressure trade-off shows that readiness is not simply a function of economic growth alone; environmental resource management must be considered alongside economic and tourism development, particularly for already-urbanised, high-tourism states. Meanwhile, Cluster 2's combination of rich natural resources but low socioeconomic readiness (Kelantan, Sabah, Sarawak) suggests these states represent priority targets for balanced, sustainability-oriented tourism investment — converting environmental assets into readiness gains without repeating Cluster 0's resource-pressure pattern.

OUTPUT (DASHBOARD)

The analytical findings were operationalised into an interactive three-page Power BI dashboard, translating multidimensional statistical outputs into an accessible decision-support tool for policymakers, tourism agencies, and researchers.

Page 1 – Demand Forecast shows national tourism trends via three KPI cards (Latest Actual Arrivals: 3.10M, Next Month Forecast: 2.36M, YoY Growth: -0.16%, color-coded red/green for direction) and a line chart comparing actual monthly arrivals against a seasonal-baseline forecast, sourced from data.gov.my's arrivals-by-state-of-entry data. The near-zero arrivals in 2020–2021 are annotated as resulting from COVID-19 border closures, giving immediate context to what would otherwise appear anomalous.

Page 2 – State Tourism Pressure presents state-level KPIs (Average Occupancy Rate: 46.26%, Total International Guests: 14M, Tourism Intensity: 440.82 guests per capita), a ranked occupancy-rate bar chart, an international-guest-volume bar chart, and a bubble scatter plot of tourism intensity vs. occupancy rate (bubble size = guest volume, each state labelled directly for clarity). A state slicer enables filtering all visuals to a single state.

Page 3 – Sustainability Analysis displays the four composite readiness dimensions (Economic: 34.29/100, Social: 55.34/100, Environment: 60.57/100, Readiness: 46.91/100) alongside contextual charts on state-level score breakdowns, treated water access, and Gini/poverty indicators. A machine-learning section shows Random Forest feature importance (led by poverty_score, inequality_score, labour_participation_rate), Model Overview cards (3 Clusters, 16 States, Silhouette Score 0.31), and a Cluster Profile Summary table. State and cluster slicers enable cross-filtering across the page.

Across all pages, the dashboard applies a consistent teal-navy-orange theme, an auto-highlighting page navigator, Report Page Tooltips (hover over the "i" icon on each chart) for guided interpretation, and configured cross-filtering. Units and formatting are applied consistently (percentages, /100 index scores, abbreviated large numbers), with clear titles, legends, and axis labels throughout — ensuring the dashboard is coherent, professional, and supports the project's decision-support goals.

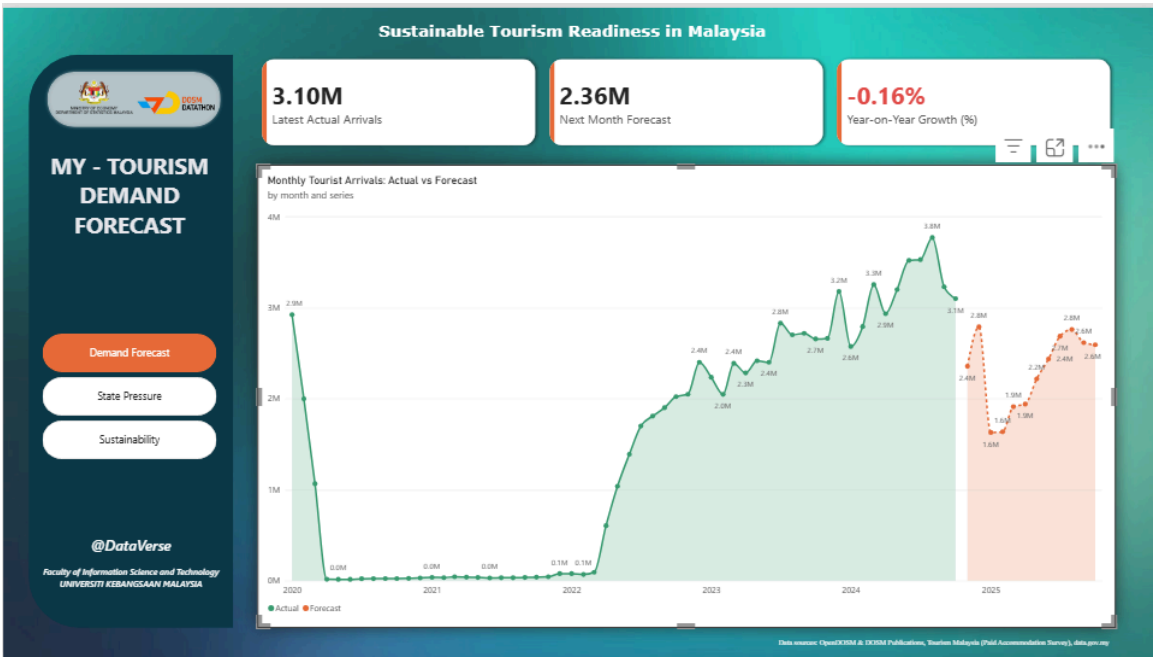


Figure 9 *Page 1: MY - Tourism Demand Forecast*

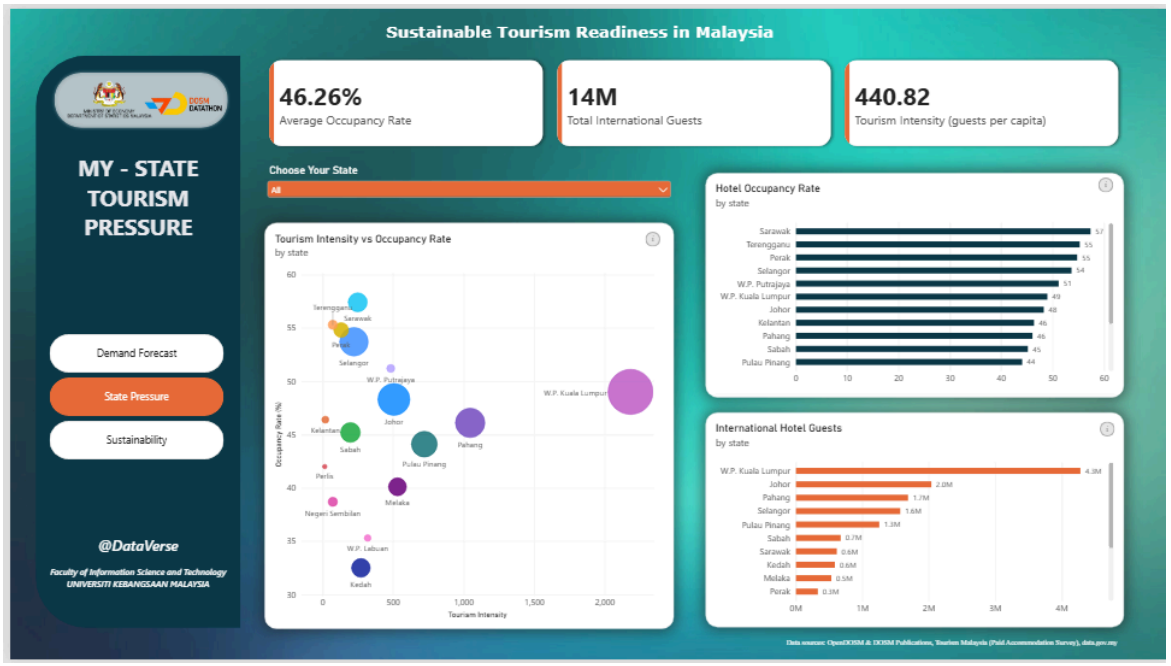


Figure 10 *Page 2: MY - State Tourism Pressure*

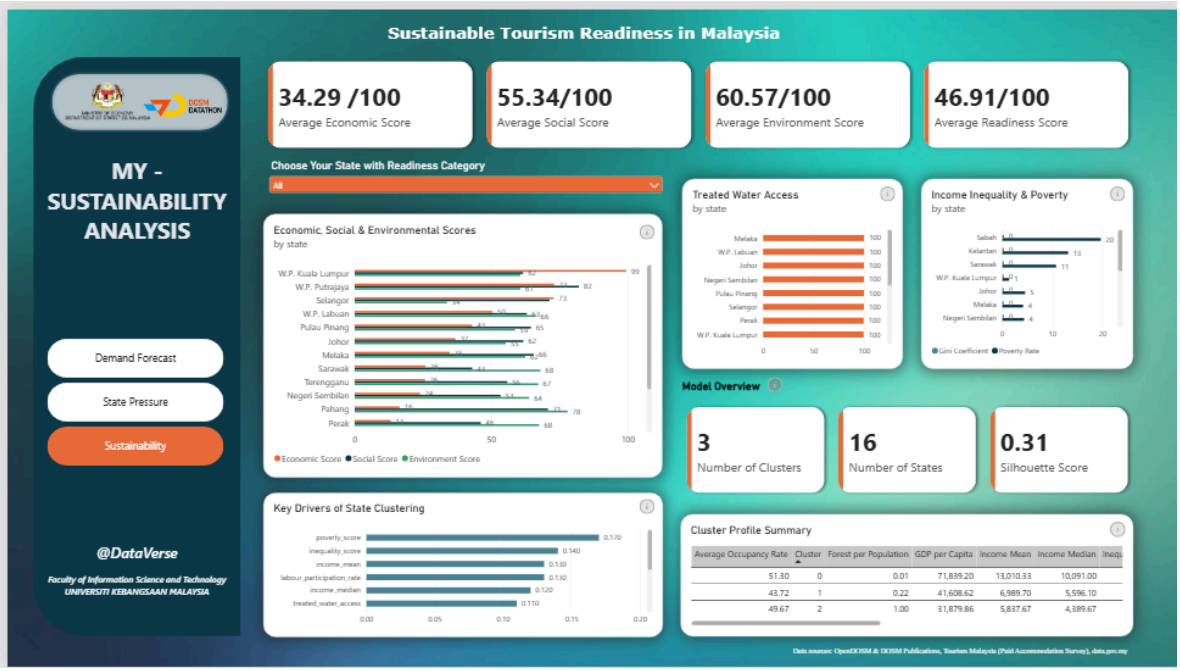


Figure 11 *Page 3: MY - Sustainability Analysis*

CONCLUSIONS

This study developed an AI-driven Sustainable Tourism Readiness Framework to evaluate tourism sustainability across Malaysian states using official datasets obtained from the OpenDOSM platform and the Department of Statistics Malaysia (DOSM). By integrating tourism, economic, social, and environmental indicators into a unified analytical framework, the project addressed the need for a more comprehensive approach to understanding tourism readiness beyond traditional tourism performance metrics. The framework was further enhanced through machine learning techniques and interactive dashboard visualisation to support evidence-based decision-making.

The findings demonstrate that sustainable tourism readiness is influenced by a combination of socioeconomic, environmental, and tourism-related factors rather than tourism activity alone. The K-Means clustering model successfully identified distinct groups of Malaysian states with similar tourism readiness characteristics, while Random Forest analysis revealed that poverty levels, household income, income inequality, treated water access, and labour participation rates were among the most influential factors affecting readiness outcomes. These findings highlight the importance of balanced development across economic, social, and environmental dimensions in

supporting sustainable tourism growth.

The developed Power BI dashboard transforms complex analytical outputs into an accessible decision-support platform through three integrated modules: Tourism Demand Forecast, State Tourism Pressure Analysis, and Sustainability Analysis. The dashboard enables policymakers, tourism agencies, and stakeholders to compare state performance, monitor tourism pressure, explore sustainability indicators, and identify priority intervention areas. As a result, decision-making can be supported by data-driven insights rather than relying solely on conventional tourism statistics.

From a practical perspective, the proposed framework has significant implementation potential within government agencies, tourism boards, local authorities, and regional planning organisations. The readiness scores and cluster classifications can be utilised to support tourism planning, resource allocation, infrastructure investment, and sustainability monitoring. States with lower readiness levels can be identified early, enabling targeted policy interventions and strategic development initiatives.

The project also contributes directly to the Sustainable Development Goals (SDGs) highlighted by DOSM Datathon 2026, particularly SDG 8 (Decent Work and Economic Growth) and SDG 12 (Responsible Consumption and Production). Through the integration of official statistics, machine learning, and interactive analytics, the proposed framework supports the development of a more sustainable, resilient, and inclusive tourism industry in Malaysia.

Future enhancements may include the incorporation of additional environmental indicators, real-time tourism data, geospatial analytics, and advanced forecasting models to further improve analytical accuracy and decision-support capabilities. The framework can also be expanded to district-level analysis and integrated into a national tourism monitoring system, allowing continuous assessment of tourism sustainability across Malaysia.

In conclusion, this project demonstrates how machine learning and official statistical data can be leveraged to create an innovative and practical decision-support solution for sustainable tourism development. By providing a comprehensive assessment of state-level readiness and identifying the factors that drive sustainable tourism outcomes, the framework offers a scalable model that can support policymakers and stakeholders in building a more sustainable tourism future for Malaysia.

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Here's the complete reference list in APA 7th edition format, alphabetized:

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